bigbasket
India's largest online supermarket



BIG BASKET

PRODUCT & PRICING

ANALYSIS

EDA with Big Basket Data

E-commerce (electronic commerce) is the activity of electronically buying or selling of products on online services or over the Internet. E-commerce draws on technologies such as mobile commerce, electronic funds transfer, supply chain management, Internet marketing, online transaction processing, electronic data interchange (EDI), inventory management systems, and automated data collection systems.

Big basket is the largest online grocery supermarket in India. It was launched somewhere around 2011 and since then they have been expanding their business. Though some new competitors have set foot in the nation, such as Blinkit, Big Basket has still not lost anything - thanks to its ever expanding popular base and the shift to online buying.

In this mini project we perform Exploratory Data Analysis (EDA) on the Big Basket products dataset using Python (NumPy, Pandas, Matplotlib and Seaborn) to understand product listings,pricing, discounts and customer ratings.

About Dataset

This dataset contains 10 attributes with simple meaning and which are described as follows:

- ***index - Simply the Index!***
- ***product - Title of the product***
- ***category - Category into which product has been classified***
- ***sub_category - Subcategory into which product has been kept • brand - Brand of the product***
- ***sale_price - Price at which product is being sold on the site***
- ***market_price - Market price of the product***
- ***type - Type into which product falls***
- ***rating - Rating the product has got from its consumers***
- ***description - Description of the product (in detail)***

Dataset size: **27,555 rowsand 10 columns (Range Index 0 to 27554).**

Steps to Follow

Step 1: Load DataSet.

Step 2: Use head function to look for first 12 rows.

Step 3: Get Description of the data in the DataFrame.

Step 4: Find Information about the DataFrame.

Step 5: Find out Top & least sold products.

Step 6: Measuring discount on a certain item.

Step 7: Find out the Missing Values from the Dataset.

Step 8: Find out the outliers from the dataset according to the columns and fill them with the mean.

Step 9: Create Plots or visualizations.

Libraries used: ***NumPy, Pandas, Matplotlib, Seaborn***

Step 1: Load DataSet

The first step ofany EDA is to import the required librariesand load the dataset into a Pandas DataFrame. We use pd.read_csv() to read the BigBasket file. The dataset has 27,555 rows and10 columns.

Python Code:

```
#Importes     import
numpy as np import
pandas as pd
import matplotlib.pyplot as plt
import seabornassns
       thedataset

# Load
df = pd.read_csv("BigBasket_Products.csv")

print(df.shape)
# Output: (27555,10)
```

Output:

(27555,10) - the dataset was loaded successfully.

Step 2: First 12 Rows using head()

The head() function shows the first rows of the DataFrame. Passing 12gives a quick look at the product names, categories, brands, prices and ratings. (Thelong description column is left outhere so the table fits on the page.)

Python Code:

```
# head(12) shows the first 12 rows
df.drop(columns="description").head(12)
```

Output:

index	product	category	sub_category	brand	sale_price	market_price	type	rating
1	Garlic Oil - Vegetarian Capsule 50	Beauty & Hygiene	Hair Care	Sri Sri Ayurveda	220.0	220.0	Hair Oil & Serum	4.1
2	Water Bottle - Orange	Kitchen, Garden & Pets	Storage & Accessories	Mastercook	180.0	180.0	Water & Fridge Bottles	2.3
3	Brass Angle Deep - Plain, No.2	Cleaning & Household	Pooja Needs	Trm	119.0	250.0	Lamp & Lamp Oil	3.4
4	Cereal Flip Lid Container/Storage	Cleaning & Household	Bins & Bathroom Ware	Nakoda	149.0	176.0	Laundry, Storage Baskets	3.7
5	Creme Soft Soap - For Hands & Body	Beauty & Hygiene	Bath & Hand Wash	Nivea	162.0	162.0	Bathing Bars & Soaps	4.4
6	Germ - Removal Multipurpose Wipes	Cleaning & Household	All Purpose Cleaners	Nature Protect	169.0	199.0	Disinfectant Spray & Cleaners	3.3
7	Multani Mati	Beauty & Hygiene	Skin Care	Satinance	58.0	58.0	Face Care	3.6
8	Hand Sanitizer - 70% Alcohol Base	Beauty & Hygiene	Bath & Hand Wash	Bionova	250.0	250.0	Hand Wash & Sanitizers	4.0
9	Biotin & Collagen Volumizing Hair	Beauty & Hygiene	Hair Care	StBotanica	1098.0	1098.0	Shampoo & Conditioner	3.5
10	Scrub Pad - Anti- Bacterial, Regul	Cleaning & Household	Mops, Brushes & Scrubs	Scotch brite	20.0	20.0	Utensil Scrub-Pad, Glove	4.3
11	Wheat Grass Powder - Raw	Gourmet & World Food	Cooking & Baking Needs	NUTRASHIL	261.0	290.0	Flours & Pre-Mixes	4.0
12	Butter Cookies Gold Collection	Gourmet & World Food	Chocolates & Biscuits	Sapphire	600.0	600.0	Luxury Chocolates, Gifts	2.2

Figure 1: Output of df.head(12)

Step 3: Description of the Data using describe()

describe() gives the statistical summary (count, mean, std, min, quartiles, max)ofall numerical columns. Note that it only shows numerical columns.

Here the averagesale price is about Rs. 334.65 against an average market price of Rs. 382.06, and theaverage rating is about 3.94 out of 5. The maximum sale price (Rs. 112,475) is veryfar from the 75th percentile (Rs. 359), which hints at outliers.

Python Code:

```
# describe() will show only numerical columns
df.describe()
```

Output:

	index	sale_price	market_price	rating
count	27555.0	27549.0	27555.0	18919.0
mean	13778.0	334.648	382.057	3.943
std	7954.588	1202.102	581.731	0.739
min	1.0	2.45	3.0	1.0
25%	6889.5	95.0	100.0	3.7
50%	13778.0	190.32	220.0	4.1
75%	20666.5	359.0	425.0	4.3
max	27555.0	112475.0	12500.0	5.0

Figure 2: Output of df.describe()

Step 4: Information about the DataFrame using info()

info() shows the index range, column names, non-null counts and data types. It is the fastest way to spot missing values and wrong data types.

Python Code:

```
# Information about the DataFrame
df.info()
```

Output:

```
<class 'pandas.DataFrame'>
RangeIndex: 27555 entries, 0 to 27554
Data columns (total 10 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   index           27555 non-null  int64
 1   product         27554 non-null  str
 2   category        27555 non-null  str
 3   sub_category    27555 non-null  str
 4   brand           27554 non-null  str
 5   sale_price      27549 non-null  float64
 6   market_price    27555 non-null  float64
 7   type            27555 non-null  str
 8   rating          18919 non-null  float64
 9   description     27440 non-null  str
dtypes: float64(3), int64(1), str(6)
```

Observation: rating has only 18,919 non-null values, so a large part of the products have no rating.

Step 5: Top & Least Sold Products

The dataset does not contain a units-sold column, so "sold" cannotbemeasured directly. As a practical proxy we use (a) the number of products listedper brand, to find the most and least prominent brands, and (b) the customer rating,tofind the best and worst rated products.

Fresho (638), bb Royal (539) and BB Home (428) have the most listings. Out of 2,313 brands, 505 brands have only a single product listed. Beauty & Hygiene is the biggest category with 7,867 products.

Python Code:

```
# Top 10 and least 10 brands by number of products
brand_cnt  =  df["brand"].str.strip().value_counts()
top_brands = brand_cnt.head(10)
least_brands = brand_cnt.tail(10)

# Top and least rated products
rated = df.dropna(subset=["rating"])
top_rated = rated.sort_values("rating", ascending=False).head(10)
low_rated = rated.sort_values("rating").head(10)

# Bar plots
sns.barplot(x=top_brands.values, y=top_brands.index)
sns.barplot(x=df["category"].value_counts().values,
            y=df["category"].value_counts().index)
```

Output:

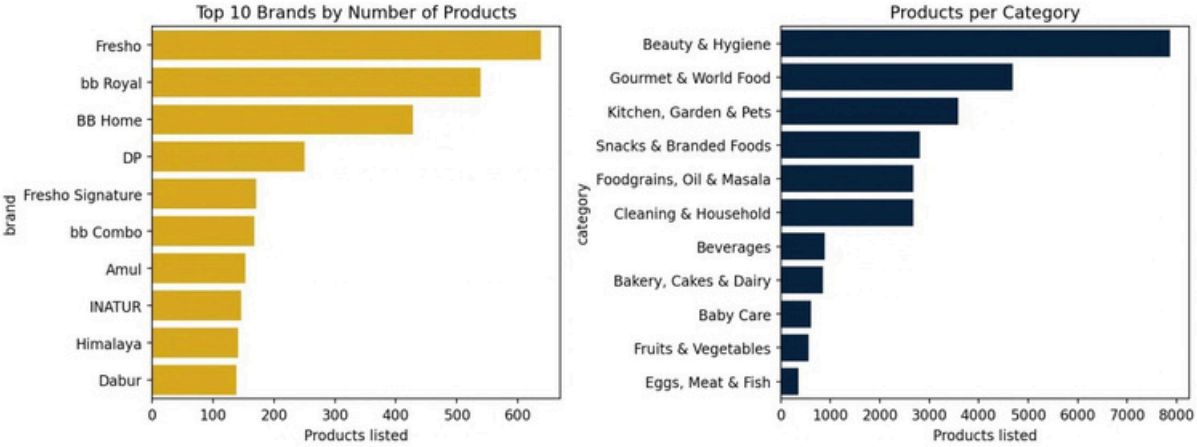


Figure 3: Top 10 brands and products per category

brand	products_listed
Nonandrai Bholanath	1
SEMCO	1
Double Pagoda	1
Muscleblaze	1
Goldrops	1
Alkalen	1
Mr. White	1
Solo	1
365	1
Baron De Bercy	1

Figure 4: Least 10 brands by number of products

product	brand	rating
Arrabbiata Tomato Pasta Sauce With	Montanini	5.0
Good Girl Eau De Parfum For Women	Carolina Herrera	5.0
Casserole Set - Die-Cast, Inductio	Wonderchef	5.0
Eau De Toilette For Men	Bentley	5.0
Pro Expert Nutrition Large Breed P	Pedigree	5.0
Olive Oil - Refined Pomace Mild	Basso	5.0
Induction Base Pressure Cooker IFP	Hawkins Futura	5.0
Steel Kitchen Fruits and Vegetable	HAZEL	5.0
Stainless Steel Dinner Set - Apple	Kitchen Essentials	5.0
Raw Manuka Honey K Factor - 16+	Wedderspoon	5.0

Figure 5: Top 10 rated products

product	brand	rating
Water/Juice Glass - Mario	Lyra	1.0
Red Chilli Sauce	Navya Food Art	1.0
Urban Balance Daily Scrub Cleanser	Faces	1.0
Charlie - Gold Edt	Revlon	1.0
Body Deodorant - Hydra Energy	Wild Stone	1.0
All In Wash	QRAA	1.0
Rosehip Pure Coldpressed Oil	StBotanica	1.0
Caramel Chocolate Bar	Rage	1.0
Whole Wheat Cake - Chocolate Marbl	BHealthy	1.0
Shikanji Syrup	Delight Foods	1.0

Figure 6: Least 10 rated products

Step 6: Measuring Discount on a Certain Item

Discountisthedifference between the market price and the sale price. It is alsouseful to expressitasapercentage: (market_price - sale_price) / market_price x 100.We first calculateitforachosen item ("Brass Angle Deep") and then for all products.

Five rowswherethesale price was greater than the market price (data entry errors) and sixrowswithamissing sale price were excluded. About 55% of the products are sold atadiscountand the average discount is about 11.8%.

PythonCode:

```
# Discount = market_price - sale_price
d = df.dropna(subset=["sale_price"]).copy()
d = d[d["sale_price"] <= d["market_price"]]
d["discount"] = d["market_price"] - d["sale_price"]
d["discount_pct"] = (d["discount"] / d["market_price"] * 100).round(2)


# Discount on a certain item
d[d["product"].str.contains("Brass Angle Deep")]\
  [["product","sale_price","market_price","discount","discount_pct"]].head(5)


# Top 10 discounted products
d.sort_values("discount_pct", ascending=False).head(10)
```

Output:

product	sale_price	market_price	discount	discount_pct
Brass Angle Deep - Plain, No.2	119.0	250.0	131.0	52.4
Brass Angle Deep Stand - Plain, No	119.0	250.0	131.0	52.4
Brass Angle Deep Stand - Plain, No	129.0	295.0	166.0	56.27
Brass Angle Deep Stand - Plain, No	165.0	375.0	210.0	56.0

Figure 7: Discount on the item "Brass Angle Deep"

product	sale_price	market_price	discount_pct
Curry Leaves	2.45	15.0	83.67
Fruit & Vegetables Hand Juicer	349.0	1995.0	82.51
Small Silicone Spatula With Plasti	75.0	399.0	81.2
Decorative Party Light Big Star St	209.0	1099.0	80.98
NHS 860 Temperature Control Profes	467.03	2395.0	80.5
Concealer Brush 930	350.0	1750.0	80.0
Decorative Party Light Golden Bell	415.0	1999.0	79.24
Decorative Party Light Golden Bell	415.0	1999.0	79.24
USB String Fairy Lights 3M 30 LED	85.0	399.0	78.7
Steel Belly Shape Storage Dabba/ C	219.0	995.0	77.99

Figure 8: Top 10 products by discount percentage

Result: "Brass Angle Deep-Plain, No.2" is sold at Rs. 119 against a market price of Rs. 250 - a discount of Rs.131 (52.4%).

Step 7: Missing Values in the Dataset

isnull().sum() counts the missing values in each column. rating is the column with the most missing values (8,636, i.e. 31.34%), followed by description (115), sale_price (6), and one each inproduct and brand.

Python Code:

```
# Missing values in each column
df.isnull().sum()

# Percentage of missing values
(df.isnull().sum() / len(df) * 100).round(2)

# Fill missing numeric values with the mean
df["rating"] = df["rating"].fillna(df["rating"].mean())
df["sale_price"] = df["sale_price"].fillna(df["sale_price"].mean())
```

Output:

column	missing_values	missing_%
index	0	0.0
product	1	0.0
category	0	0.0
sub_category	0	0.0
brand	1	0.0
sale_price	6	0.02
market_price	0	0.0
type	0	0.0
rating	8636	31.34
description	115	0.42

Figure 9: Missing values per column

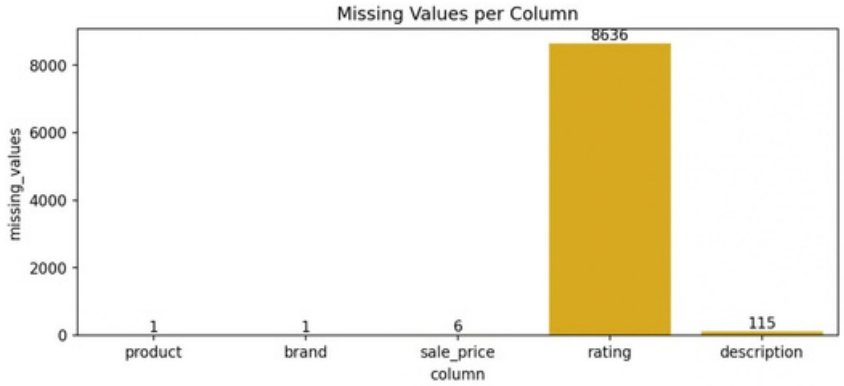


Figure 10: Bar chart of missing values

Step 8: Outliers and Filling them with the Mean

Outliers are detected using the IQR method: any value below Q1 - 1.5 x IQRorabove Q3 + 1.5 x IQR is treated as an outlier. It is applied to sale_price and market_price. The outliers are then replaced with the mean of the remaining (non-outlier) values. sale_price has 2,205 outliers (upper limit Rs. 755) and market_price has 2,147outliers (upper limit Rs. 912.5). The box plots below show the data before and after the treatment.

Python Code:

```
for col in["sale_price", "market_price"]:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR =Q3-    Q1
    lower=Q1-     1.5 * IQR
    upper=Q3+1.5 * IQR
    mask =(df[col] < lower) | (df[col] > upper)
    print(col,"outliers:", mask.sum())
    # filloutliers with mean of non-outlier values
    df.loc[mask,col] = df.loc[~mask, col].mean()

sns.boxplot(x=df["sale_price"])
```

Output:

column	Q1	Q3	IQR	lower	upper	outliers	fill_mean
sale_price	95.0	359.0	264.0	-301.0	755.0	2205	217.84
market_price	100.0	425.0	325.0	-387.5	912.5	2147	255.62

Figure 11: IQR limits, outlier count and fill value

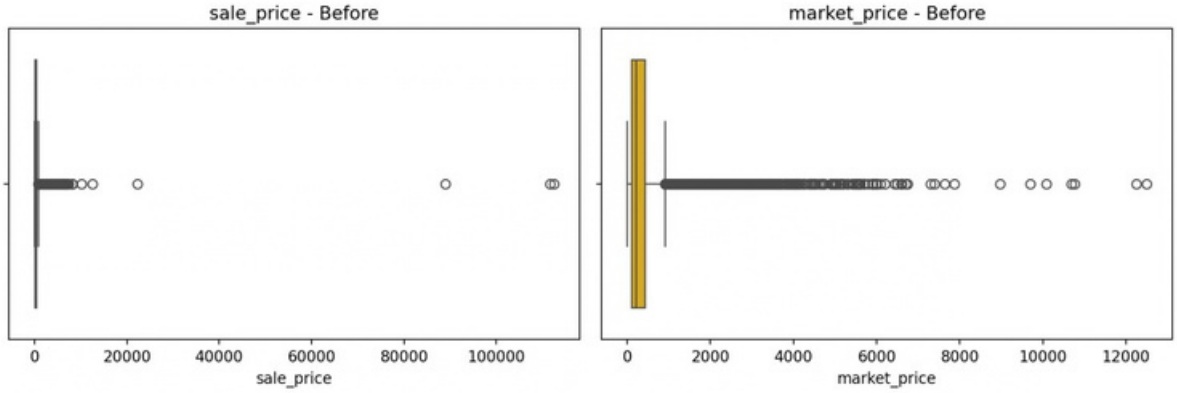


Figure 12: Box plots BEFORE treating outliers

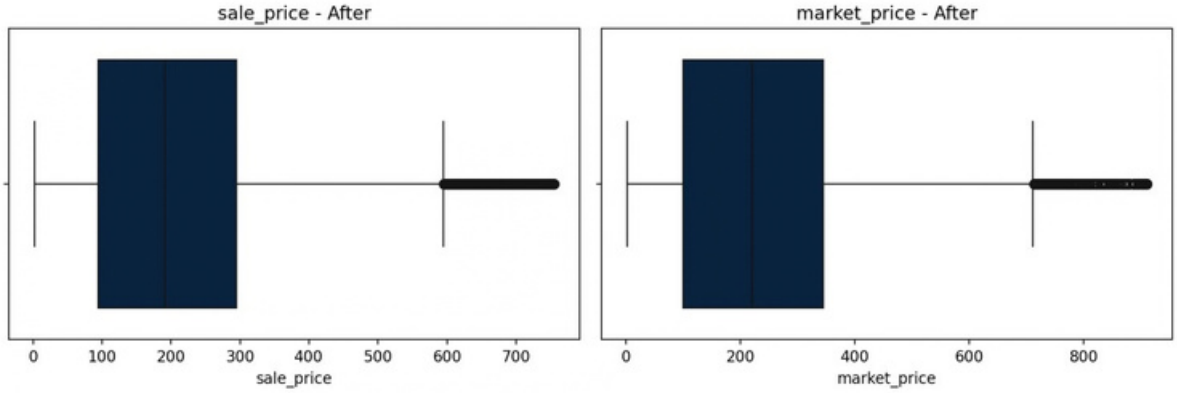


Figure 13: Box plots AFTER filling outliers with the mean

Step 9: Plots and Visualizations

Finally, visualizations are created to understand the data better. Thecharts below show how ratings are distributed, how products are spread across categories, how sale price relates to market price, which categories give the highest discounts,and how the numerical columns are correlated.

Python Code:

```
# Rating distribution
sns.histplot(df["rating"], bins=20, kde=True)


# Category share
df["category"].value_counts().plot.pie(autopct="%1.1f%%")


# Sale price vs market price
sns.scatterplot(x="market_price", y="sale_price", data=df.sample(3000))


# Average discount % by category
d.groupby("category")["discount_pct"].mean().sort_values().plot.barh()


# Rating by category and correlation heatmap
sns.boxplot(x="rating", y="category", data=df)
sns.heatmap(df[["sale_price","market_price","rating"]].corr(), annot=True)
```

Output:

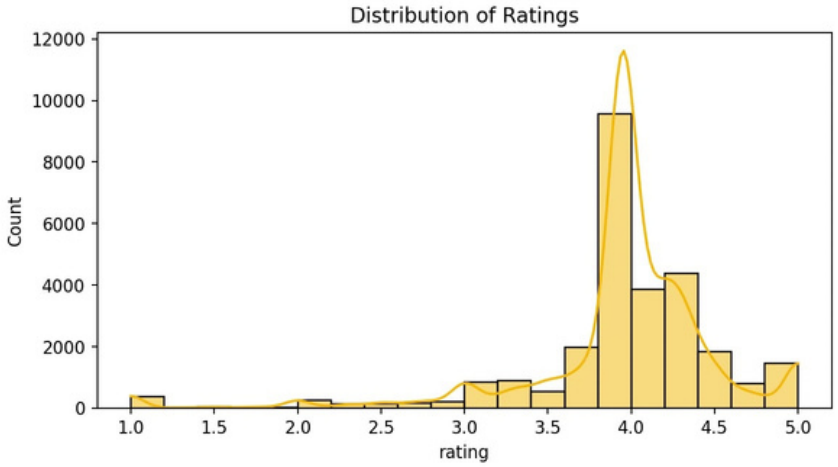


Figure 14: Distribution of ratings

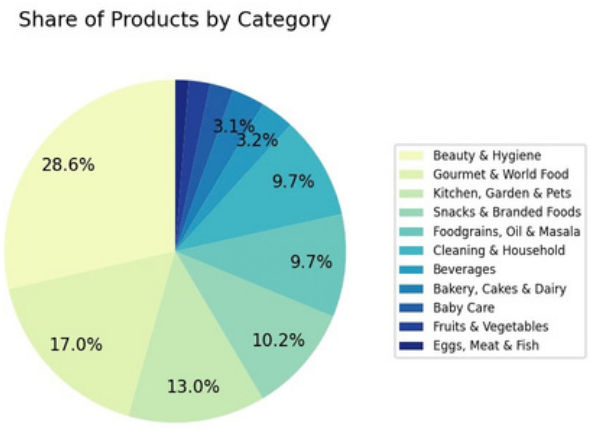


Figure 15: Share of products by category



Figure 16: Sale price vs market price

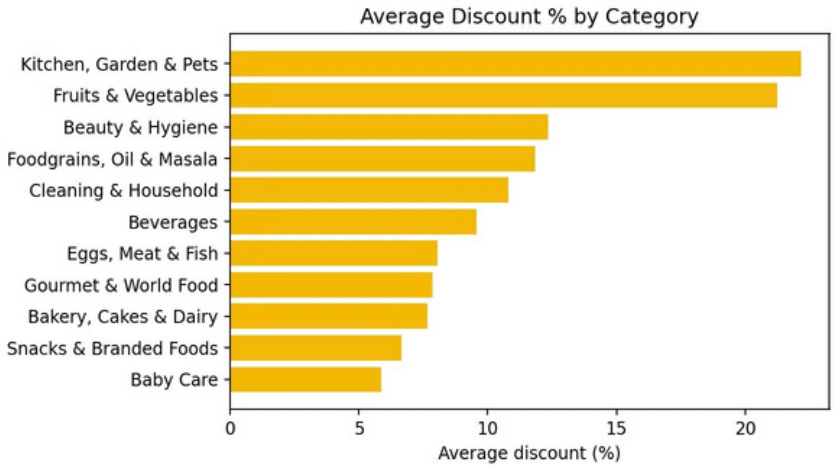


Figure 17: Average discount percentage by category

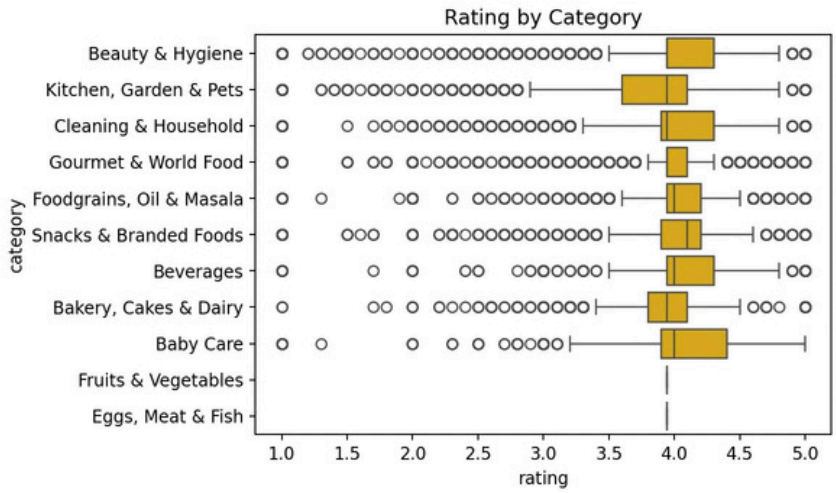


Figure 18: Rating by category

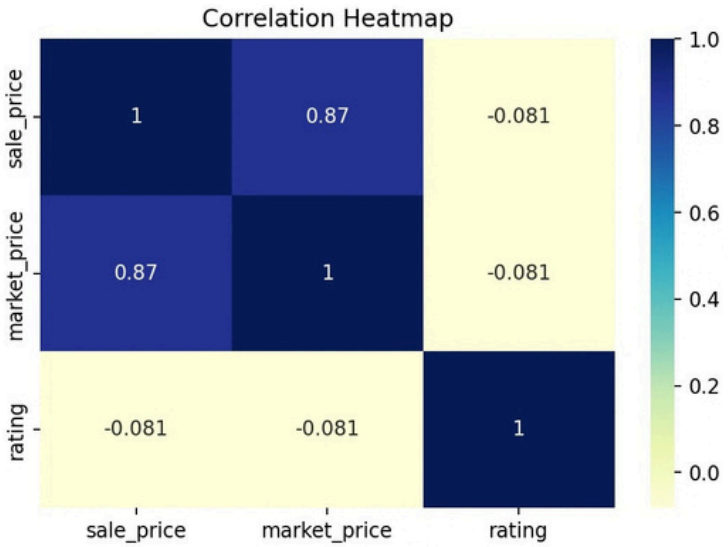


Figure 19: Correlation heatmap

Key Insights & Conclusion

- **Brand Analysis:** Fresho, bb Royal and BB Home have themost listings; 505 of the 2,313 brands have just one product.
- **Category Analysis:** Beauty & Hygiene is the largest category (7,867 products, about 28.6%), followed by Gourmet & World Food and Kitchen, Garden & Pets.
- **Ratings:** Ratings are mostly between 3.7 and 4.3 (average 3.94), but 31.34% of products have no rating.
- **Discounts:** About 55% of products are discounted, with an average discount of about 11.8%. Kitchen, Garden & Pets (22.2%) and Fruits & Vegetables (21.3%) have the highest average discounts.
- **Pricing:** Sale price and market price are strongly correlated (0.87), while rating has almost no relationship with price (-0.08).
- **Data quality:** Missing values and price outliers were found and treated; a few rows with sale price above market price should be verified at source.

Conclusion: The exploration gives a clear view of Big Basket's product mix, pricing and discount strategy. Cleaning the data with Pandas (missing values and outliers) and visualizing it with Matplotlib and Seaborn helped to answer the business queries and to understand which categories and brands drive the catalogue.